

Parent-Level Debt and Subsidiary-Level Risk-Taking: Evidence from Large Bank Holding Companies *

Kebin Ma[†] David R. Skeie[‡] Jianlin Wang[§] Jing Ye[¶]

September 24, 2026

Abstract

We investigate the influence of the parent company’s capital structure of a bank holding company’s (BHC) on the risk-taking behavior of its subsidiary banks, taking into account the BHC’s organizational structure. Our analysis shows that increased parent debt is associated with reduced risk-taking by subsidiary banks. This effect is particularly prominent in BHCs with more than one subsidiary bank or significant nonbank business, which would give the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank’s risk profile.

Keywords: bank holding company, parent-level debt, bank subsidiary, risk management

JEL Codes: G21, G28, G32, G33, G38

*We thank Yun Wang (CTFM discussant), Rebel A. Cole (AFBC discussant), and seminar participants at the 38th Australasian Finance and Banking Conference (2025), Contemporary Topics on Financial Markets (2025), and the 2024 Warwick Business School Brown Bag Series for helpful comments and feedback at various stages of this project. We also thank the staff at the Federal Reserve Banks across different districts for providing the FR Y-6 Annual Report of Holding Companies.

[†]Warwick Business School, University of Warwick. E-mail: kebin.ma@wbs.ac.uk

[‡]Warwick Business School, University of Warwick. E-mail: david.skeie@wbs.ac.uk

[§]Leavey School of Business, Santa Clara University. E-mail: jianlin.wang@scu.edu

[¶]Warwick Business School, University of Warwick. E-mail: jing.ye@warwick.ac.uk

1 Introduction

Since [Jensen and Meckling \[1976\]](#), corporate finance has a long tradition of considering the incentive implications of capital structure; the study of commercial banks makes no exception. Numerous empirical studies examine the impact of banks' capital structure on their risk-taking incentives, and virtually all studies explaining bank risk-taking would include bank capitalization as a control variable. Yet most modern commercial banks operate within a bank holding company (BHC), raising an essential question: which entity's capital position matters? A BHC can have several relevant balance sheets: the commercial bank's own, those of other subsidiaries, the parent company's, and the consolidated balance sheet of the whole group's. While prior research typically focuses on the capitalization of the commercial bank *or* that of the consolidated BHC, this paper argues that one needs to take a wholistic view, emphasizing the capital position of the parent company. Specifically, we study how the *parent debt ratio*, defined as the share of a BHC's non-deposit debt issued by its parent company, affects the risk-taking of its commercial bank subsidiaries.

[Avraham et al. \[2012\]](#) provide an insightful overview of the organizational structure of US bank holding companies. Figure 1 reproduces their findings for a typical large BHC, in which the bank holding company simultaneously controls a bank subsidiary, a non-bank subsidiary, and a commercial bank.¹ This multi-tiered organizational structure corresponds to a multi-tiered capital structure, as proxied by the allocation of external debt. The effects of parent-level debt on subsidiary bank behavior remain unexplored. We address this gap by examining how the parent debt ratio influences subsidiary banks'

¹Some BHCs also own foreign bank and non-bank subsidiaries.

risk-taking—an important question given that commercial banks account for the bulk of BHC assets, rising from 89.03% in 2012Q1 to 94.45% in 2022Q4 (Figure 2).

We find that a higher parent debt ratio is associated with decreased risk-taking by subsidiary banks. Specifically, an increase in the parent debt ratio correlates with lower loan yield, lower provision ratio, and a higher subsidiary bank Z-Score. These findings suggest that higher levels of parent debt ratio can lead to reduced risk-taking within subsidiary banks.

The underlying mechanism can be described as follows. When the parent company raises external debt and channels the proceeds into its subsidiary banks as equity, the resulting increase in capital constrains the subsidiary banks’ incentives to take on risk, since potential losses are more fully absorbed by this equity buffer. Therefore, this parent-level debt effectively serves as a contractual commitment that helps mitigate risk-taking at the subsidiary level. In contrast, if the parent company issues less debt, subsidiary banks may need to rely more heavily on their own debt issuance to finance investments, which can encourage greater risk-taking due to limited liability. Under limited liability, downside losses are partly borne by debt holders, while shareholders retain the upside gains.

Similar reasoning extends to the case where the parent issues little debt but there is cross-guarantee. The cross-guarantee provision of Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA) permits the Federal Deposit Insurance Corporation (FDIC) to charge off any expected losses related to the failure of one subsidiary bank of a multi-bank holding company to the capital of a related subsidiary bank within the same multi-bank holding company ([Ashcraft \[2005\]](#)). This implies that BHCs with multiple subsidiary banks may engage in less risk-taking, as they must use their own capital to rescue failing subsidiary banks before FDIC intervention. Our findings

confirm that subsidiary banks affiliated with multi-bank BHCs take on less risk when the parent company issues more debt.

Our paper highlights the function of internal capital markets within BHCs. Unlike existing studies—some of which focus on risk-sharing through internal capital markets among global banks (e.g., [Cetorelli and Goldberg \[2012\]](#))—we emphasize the role of internal capital markets in shaping the risk-taking incentives of domestic subsidiary banks within U.S. BHCs.

While the empirical evidence is suggestive, a key concern in identifying the effect of parent debt on a subsidiary bank’s risk-taking is omitted variable bias. Specifically, when a subsidiary bank faces stringent regulatory capital requirements, it may reduce its risk-taking to avoid potential capital losses. At the same time, the parent company may issue additional external debt and inject the proceeds into the subsidiary bank to help it meet these requirements. In this case, the observed negative relationship is likely driven by the omitted factor—namely, the tightening of regulatory capital constraints—rather than a direct causal effect of parent debt.

To address this identification concern, we exploit two related regulatory shocks as a natural experiment: the introduction of the Single Point of Entry (SPOE) resolution strategy and the Total Loss-Absorbing Capacity (TLAC) rule. These shocks generated plausibly exogenous variation in bank holding companies’ (BHCs’) incentives to issue parent-level debt, variation that is unlikely to be driven by the risk-taking decisions of their subsidiary banks. This setting therefore mitigates endogeneity concerns arising from omitted variables, and allows us to estimate the causal effect of parent debt on subsidiary bank risk-taking.

As the SPOE strategy and the TLAC framework were phased in during 2014–2015, large non-GSIB BHCs likely anticipated that they would fall outside the scope of the TLAC requirement. In response, these BHCs reduced parent-level debt issuance, which in turn strengthened their subsidiary banks’ risk-taking incentives.

We implement a difference-in-differences design around the introduction of SPOE and TLAC. We compare non-GSIB BHCs with total consolidated assets of \$50 billion or more as of 2014Q1 with two groups: GSIBs, and a control group of BHCs below the \$50 billion threshold. The pre-period (2012–2013) precedes the proposal of both SPOE and TLAC, and the post-period (2016–2017) follows the confirmation of the scope of the U.S. TLAC rule. The \$50 billion threshold corresponds to the Enhanced Prudential Standards (EPS) under Section 165 of the Dodd–Frank Act, which apply to BHCs with total consolidated assets of \$50 billion or more from 2014Q1. Consistent with our hypothesis, we find that, relative to the control group, \$50 billion-plus non-GSIB BHCs increase risk-taking following the resolution of TLAC uncertainty: their subsidiary banks exhibit higher loan yields, higher provision ratios, and higher leverage, together with lower stability as measured by the Z-score. In contrast, GSIBs, for which neither SPOE nor TLAC constituted new information, exhibit no comparable change, which provides a useful placebo test for our identification strategy. Taken together, these results support a causal interpretation of the relationship between parent debt and subsidiary bank risk-taking documented in Section 4.

We present a simple model that captures our empirical finding. In the model, the BHC comprises both a subsidiary bank and a non-bank subsidiary. The parent company issues debt and allocates the proceeds as equity investments into these subsidiaries. The subsidiary bank faces investment opportunities that involve a trade-off between risk and return. our model predicts that when the parent company issues more debt and invests

the proceeds as equity investment into its subsidiaries, it can mitigate the subsidiary bank’s risk-taking behavior. Conversely, if the parent company issues less debt, the subsidiary bank needs to increase its own debt issuance to finance its investments, which can incentivize greater risk-taking. This occurs because, under limited liability, the downside is shared with debt holders, while shareholders pocket the upside.

This study contributes to the large literature on the financial ratio regulations aimed at curbing financial institutions’ risk-taking incentives. The specific ratio examined here is the parent company’s capital structure, which differs from prior research that primarily focus on capital structure at the commercial bank level or the consolidated BHC level (e.g., [Keeley \[1990\]](#), [Shrieves and Dahl \[1992\]](#), [Demsetz and Strahan \[1997\]](#), and [Ellul and Yerramilli \[2013\]](#)).

This paper also contributes to the literature on the organizational structure of financial institutions and risk-taking. Our analysis focuses on the multi-level organizational structures of BHCs and their distinct balance sheets—particularly that of the parent company—rather than the single balance sheet of integrated conglomerates emphasized by [Freixas et al. \[2007\]](#). Moreover, we highlight the importance of considering existing BHC organizational structures when assessing risk-taking, in contrast to [Kahn and Winton \[2004\]](#) and [Luciano and Wihlborg \[2018\]](#), who design optimal structures to mitigate risk-taking. Closest to our approach is [Li et al. \[2024\]](#), who argue that a bank’s organizational structure should be evaluated alongside its capital structure and bankers’ compensation contracts when discouraging banker’s risk-taking.

Finally, we add to the nascent literature on parent debt in BHCs, where existing research is scarce. [Tucker et al. \[2023\]](#) show that the parent-only leverage ratio is positively associated with the parent’s credit default swap premium, suggesting that parent-only balance sheet data can inform assessments of parent credit risk and that consolidated

reporting can obscure critical entity-level details. Building on these insights, we examine how parent company debt affects subsidiary bank risk-taking, offering a novel perspective on risk transmission within BHCs.

The paper is structured as follows: Section 2 describes the data sources, key variables, and summary statistics. Section 3 conducts the empirical analysis on the relationship between parent debt and subsidiary banks’ risk-taking. Section 4 presents the difference-in-differences analysis. Section 5 examines the impact of the cross-guarantee mechanism. Section 6 conducts robustness test. Section 7 introduces the theoretical model. Finally, Section 8 concludes.

2 Data and summary statistics

In this section, we describe our data sample, outline the construction of the parent debt ratio and subsidiary banks’ risk-taking measures, and summarize our sample. Table 1 provides detailed definition of these variables.

2.1 Data sources

2.1.1 BHC-level data

Our BHC-level data are sourced from the Consolidated Financial Statements for Holding Companies (FR-Y9C) and Parent Company Only Financial Statements for Large Holding Companies (FR-Y9LP) provided by the Federal Reserve Bank. We use data spanning from 2011 to 2022Q2, which include quarterly observations of the income statement and balance sheets of all U.S. BHCs and their parent-only companies. We drop BHCs that

do not file the FR-Y9C or FR-Y9LP for each quarter during the entire sample period to avoid the impact of BHCs' joining or exiting on our empirical results. Finally, we collect data for 298 BHCs over the sample period.

2.1.2 Subsidiary bank-level data

The subsidiary bank-level data are from the U.S. Call Reports provided by the Federal Reserve Bank, covering the period from 2011 to 2022Q2. This dataset contains quarterly observations of the income statement and balance sheets of all U.S. commercial banks, some of which are the subsidiaries of BHCs. The data contain bank-level identifier (RSSD9348) that can be used to link to the BHC-level data. Finally, our balanced panel consists of 157 unique subsidiary banks.

2.2 Parent debt ratio

Our key explanatory variable is the parent debt ratio, defined as the parent's liability scaled by the BHC's total consolidated assets. The parent's liability includes the parent company's deposits, securities sold under agreements to repurchase, borrowings of commercial paper and other borrowings with a remaining maturity of one year or less, other borrowed money with a remaining maturity of more than one year, and subordinated notes and debentures.

2.3 Subsidiary bank's risk-taking

We measure subsidiary bank risk along both the asset and liability sides of the balance sheet. Our first measure is the Z-score, defined as the sum of the return on assets (ROA)

and the equity-to-assets ratio, scaled by the standard deviation of ROA, where the equity-to-assets ratio is book equity divided by total assets. The Z-score thus captures both asset-side risk, through the level and volatility of returns, and liability-side risk, through capitalization; a higher Z-score indicates greater bank stability. Because the Z-score is highly skewed, we use its natural logarithm, which is approximately normally distributed (Laeven and Levine [2007]). Second, similar with Acharya et al. [2017], we measure leverage as the ratio of the subsidiary bank’s debt to its equity. Third, we use two measures that capture risk-taking in the loan portfolio more directly. The loan yield, defined as quarterly interest income on loans divided by quarterly average loans and annualized, is an ex ante measure: higher yields indicate lending to riskier borrowers. The provision ratio, defined as provisions for credit losses divided by quarterly average loans and leases and annualized, reflects expected credit losses in the loan portfolio (Hirtle et al. [2020]).

2.4 Control variables

We use market data from the Center for Research in Security Prices (CRSP) and regulatory accounting data from Call Reports, FR Y-9C, and FR Y-9LP to construct subsidiary bank- and BHC-level controls that are considered as important determinants of a bank’s risk-taking behavior.

These control variables include subsidiary bank size ($LnTA$), the subsidiary bank’s leverage ratio ($Debt/Equity$), high-quality liquid assets ($HQLA$), and the BHC’s Tobin’s Q ($TobinQ$). Keeley [1990] shows that Tobin Q, as a proxy of market power or charter value, is negatively related to a bank’s risk-taking. Boyd and Runkle [1993] suggest a negative relationship between bank size and bank risk. Acosta-Smith et al. [2020]

find that a leverage ratio requirement associated with the introduction of the Basel III Leverage Ratio can incentivize banks bound by it to increase their risk-taking.

2.5 Summary statistics

Table 2 summarizes the sample statistics for the 2012-2021Q2 period. For the BHC-level variables, the parent debt ratio averages 1.87%. The average Tobin’s Q for BHCs is 1.03. The mean value of HQLA ratio is 15.43%.

For the subsidiary bank-level variables, the mean logarithm of asset size is 15.02, which translates to a mean of USD 3.34 billion in total assets. Regarding subsidiary banks’ risk-taking measures, the mean logarithm of the Z-score is 3.65, the average loan yield is 4.82%, the provision ratio is 0.32%, and the mean leverage ratio is 7.29.

3 Empirical approach and results

The central premise of our paper is the impact of parent debt ratio on the risk-taking of subsidiary banks. In this section, we present evidence that establishes the negative link between the parent debt ratio and subsidiary bank’s risk-taking.

For each subsidiary bank, we estimate the following model:

$$\begin{aligned} RiskTaking_{ijt} = & \alpha + \beta ParentDebtRatio_{j,t-4} + \\ & + \gamma Controls_{i,t-4} + \delta Controls_{j,t-4} + \eta_t + \epsilon_{ijt} \end{aligned} \quad (1)$$

where, for each subsidiary bank i affiliated with BHC j and in quarter t , *ParentDebtRatio* is calculated as the parent’s liability divided by the BHC’s total assets, *Controls* represents a set of subsidiary bank-level and BHC-level control variables described in Section 2.4. The BHC-level control variables are *HQLAratio* and *TobinQ*. Since most subsidiary banks are not public, we could not calculate the bank-level Tobin Q. Therefore, we use BHC-level Tobin Q as an alternative in our regression model. All explanatory variables are lagged by four quarters to mitigate concerns over look-ahead bias. *RiskTaking* is a measure of bank’s risk-taking originating in quarter t described in Section 2.3. We include subsidiary bank and quarter fixed effects to mitigate omitted-variable bias. In all models, standard errors are clustered at subsidiary bank level.

Table 3 presents the empirical evidence showing a negative relationship between the parent debt ratio and the subsidiary bank’s risk-taking during the 2011–2021Q2 period. Specifically, for a one standard deviation increase in the parent debt ratio (0.033), we observe an expected 9.80% increase in Z-Score ($2.97 \times 0.033 \times 100\% = 9.80\%$), a 0.20% decrease in loan yield ($-0.062 \times 0.033 \times 100\% = -0.20\%$), a 0.16% decrease in provision ratio ($-0.048 \times 0.033 \times 100\% = -0.16\%$), and a 20.23% decrease in leverage ($-16.91 \times 0.033 \times 100\% = -55.80\%$).

To compare magnitudes across outcomes, we scale each of these changes by the standard deviation of the respective outcome variable. The effects are economically meaningful. For the Z-score, whose standard deviation is 0.579, a one-standard-deviation increase in the parent debt ratio (0.033) is associated with an increase of 0.17 standard deviations ($2.97 \times 0.033 / 0.579 = 0.17$). By the same measure, a one-standard-deviation increase in the parent debt ratio is associated with decreases of 0.13 standard deviations in the loan yield ($-0.062 \times 0.033 / 0.0157 = -0.13$), 0.23 standard deviations in the

provision ratio $(-0.048 \times 0.033 / 0.0068 = -0.23)$, and 0.18 standard deviations in the leverage ratio $(-16.91 \times 0.033 / 3.0585 = -0.18)$.

Taken together, these magnitudes indicate that parent-level debt meaningfully reduces subsidiary banks' risk-taking.

4 The introduction of SPOE and TLAC as a natural experiment

We begin our analysis by outlining the background of the single-point-of-entry (SPOE) resolution strategy, which is operationalized through the U.S. Total Loss-Absorbing Capacity (TLAC) framework.

4.1 Regulatory background

On December 18, 2013, the Federal Deposit Insurance Corporation (FDIC) published a notice describing the SPOE strategy for resolving systemically important financial institutions (SIFIs) under the Orderly Liquidation Authority of the Dodd–Frank Act. The strategy is designed to allow a failing SIFI to be resolved without causing serious adverse effects on U.S. financial stability. Under SPOE, only the top-tier holding company, that is, the parent company of the holding company, enters resolution, while its operating subsidiaries continue as going concerns. For the strategy to be credible, the parent company must hold a sufficiently thick layer of equity and long-term unsecured debt that can absorb losses and be used to recapitalize its subsidiaries in resolution.

The SPOE notice itself did not impose binding requirements; the subsequent TLAC rule made the strategy operational. In response to a call by the G20 Leaders at the St. Petersburg Summit in September 2013 for proposals to enhance the loss-absorbing capacity of global systemically important banks (GSIBs) in resolution, the Financial Stability Board (FSB) issued a proposed term sheet on TLAC in November 2014. The Federal Reserve proposed its own TLAC and long-term debt rule in October 2015, requiring U.S. GSIBs to maintain minimum amounts of capital and long-term unsecured debt at the parent company level.

The U.S. TLAC rule was adopted under the Federal Reserve’s enhanced prudential standards (EPS) authority in Section 165 of the Dodd–Frank Act and codified in Regulation YY. However, whereas the EPS rule finalized in February 2014 applies to all BHCs with total consolidated assets of \$50 billion or more, TLAC applies only to GSIBs. Large non-GSIB BHCs therefore share the EPS regime with GSIBs but are not subject to TLAC, so comparing the two groups allows us to separate the effect of TLAC from that of the broader enhanced prudential standards. BHCs with total consolidated assets below \$50 billion were subject to neither EPS nor TLAC, which allows us to isolate the effects of these regulations across groups.

4.2 Identification strategy

The main identification challenge is to find an exogenous source of variation in BHCs’ incentives to hold loss-absorbing debt at the parent level. We exploit the introduction of the SPOE strategy and the TLAC requirement as a natural experiment, treating 2014–2015 as a transition period characterized by regulatory uncertainty. Three institutional features of this period motivate our design.

First, the SPOE strategy was designed for systemically important financial institutions (SIFIs) and is invoked when the failure of such an institution would pose serious risks to U.S. financial stability. Because SPOE imposes losses on the parent company while keeping operating subsidiaries open, it requires the parent to hold sufficient capital and long-term debt to absorb those losses. When the FDIC released its SPOE proposal in December 2013, it was therefore unclear whether large non-GSIB BHCs would also be expected to adopt the strategy and issue additional long-term debt at the parent level.

Second, in 2014Q1, both GSIBs and large non-GSIB BHCs with total consolidated assets of \$50 billion or more became subject to the same enhanced prudential standards (EPS) under the Federal Reserve’s final rule. The EPS rule imposed neither SPOE-related requirements nor a TLAC requirement, which may have partly eased the uncertainty that large non-GSIB BHCs faced about their obligations under SPOE and TLAC.

Third, the gradual introduction of TLAC ultimately resolved the uncertainty about which BHCs would be required to implement SPOE and issue external long-term debt at the parent level. Because capital and long-term unsecured debt at the parent level are necessary for SPOE to operate, the scope of TLAC effectively determined the scope of SPOE. The FSB’s November 2014 proposed term sheet limited TLAC to GSIBs, signaling to large non-GSIB BHCs that they would not need to meet TLAC requirements or issue long-term debt at the parent level. The Federal Reserve’s October 2015 proposal then confirmed that large non-GSIB BHCs would be exempt from the U.S. TLAC rule.

We therefore define 2014–2015 as the transition period of the natural experiment. As this uncertainty dissipated over the period, we expect large non-GSIB BHCs to reduce their parent debt issuance. The pre-period is 2012–2013, before the formal proposals of

SPOE and TLAC, and the post-period is 2016–2017, after the scope of the U.S. TLAC rule had been confirmed.

This setting allows us to sort the BHCs in our sample into three groups. The treatment group consists of non-GSIB BHCs with total consolidated assets of \$50 billion or more as of 2014Q1, which we refer to as Treat-BHCs. The control group consists of BHCs with total consolidated assets below \$50 billion as of 2014Q1. The remaining BHCs are GSIBs.

Figure 3 and Figure 4 support this identification approach. Figure 3 shows that the parent debt ratios of the treatment and control groups move in parallel until 2014, when the SPOE strategy and EPS rule took effect. After 2014, the ratio for the treatment group declines sharply, whereas the ratio for the control group remains stable, with a slight upward trend. Specifically, the parent debt ratio of the treatment group decreases from 3.5% in 2014Q1 to 3.0% in 2015Q4, whereas that of the control group increases from 0.4% to 0.5%. Figure 4 shows that the parent debt ratio of GSIBs rises sharply after the U.S. TLAC rule was proposed at the end of 2015, from 9.0% in 2015Q4 to 9.6% in 2017Q3.

We first examine whether risk-taking increases more for large non-GSIB BHCs than for BHCs in the control group after the proposal of the U.S. TLAC rule.

4.3 Econometric framework: Difference-in-differences estimator

We estimate a difference-in-differences model using 2012–2013 as the pre-period and 2016–2017 as the post-period. For each BHC, we specify our baseline model as follows:

$$\begin{aligned}
RiskTaking_{ijt} = & \alpha + \beta Post_t + \gamma Treat_BHC_{ij} \times Post_t \\
& + \lambda GSIBs_{ij} \times Post_t + \delta Controls_{ijt} + \epsilon_{ijt}
\end{aligned} \tag{2}$$

where *Post* is a dummy variable equal to 1 for the 2016–2017 post-period and 0 for the 2012–2013 pre-period, and *Treat_BHC* is a dummy variable equal to 1 for large non-GSIB BHCs subject to the Enhanced Prudential Standards but not SPOE strategy or TLAC requirement. *GSIBs* is an indicator equal to 1 for GSIBs. *Controls* is a set of subsidiary bank- and BHC-level control variables. α_i denotes subsidiary bank fixed effects. The model does not include the stand-alone *Treat_BHC* dummy variable, because it is subsumed by the subsidiary bank-specific fixed effect α_i . *RiskTaking* denotes one of our measures of subsidiary bank risk-taking: the loan yield, the provision ratio, subsidiary bank leverage, or the Z-score.

We exclude 2014–2015 from the estimation sample, treating it as the transition period during which the scope of SPOE and TLAC was resolved and parent-level debt structures were adjusted. This follows common practice in difference-in-differences designs with a gradually revealed shock, in which observations recorded while the shock is still unfolding are excluded to avoid attenuating the estimated treatment effect.

We include the interaction $GSIBs \times Post$ alongside our coefficient of interest, $Treat_BHC \times Post$, for the following reason. It serves as a placebo test of our identifying assumption. GSIBs were subject to SPOE and EPS from the outset, and their TLAC obligation never depended on how the scope of the rule was resolved during the drafting period. If our identification is valid, the subsidiary banks of GSIBs should exhibit no discernible shift in risk-taking over the same window, and λ should be statistically indistinguishable

from zero. A significant λ would instead suggest that a confounding factor common to all large BHCs, rather than the resolution of TLAC uncertainty specific to non-GSIB BHCs, drives our results.

To address the downward bias in standard errors that arises in difference-in-differences estimation with serially correlated outcomes, we follow [Bertrand et al. \[2004\]](#) and take the averages of the observations in our sample in the pre-period (2012–2013) and in the post-period (2016–2017). We will conduct a range of robustness checks and falsification tests, including alternative sample periods, to verify our results and rule out alternative explanations. In one such test, we use the transition period (2014–2015) as the post-period to illustrate the gradual decline in the parent debt ratio shown in Figure 3.

Table 4 summarizes the collapsed pre- and post-period sample of the treatment group, and Table 5 reports the regression results. Cross all risk-taking measures, relative to BHCs below the \$50 billion threshold, \$50 billion-plus non-GSIB BHCs ($Treat_BHC \times Post$) increase risk-taking following the resolution of TLAC uncertainty, over and above the effect of the preceding EPS shock, relative to BHCs below the \$50 billion threshold. In contrast, GSIBs ($GSIB \times Post$) exhibit no significant change, which is precisely the pattern predicted by our identification strategy. The coefficient for the treatment effect ($GSIB \times Post$) is statistically significant at the 1% level. Model (1) of Table 4 shows that, on average, large non-GSIB BHCs experienced an increase in loan yield that was 0.006 greater than that of BHCs in the control group, whose loan yield declined modestly over the same period. This effect is robust to the inclusion of bank-level controls in Model (2). Similarly, relative to the control group, large non-GSIB BHCs experienced an average increase of 0.003 in the provision ratio (Model (4)) and of 0.791 in the leverage ratio (Model (12)). Model (10) further shows that large non-GSIB BHCs experienced a decline in the Z-score that was, on average, 0.236 greater than that of the

control group. Collectively, these results indicate that large non-GSIB BHCs increased their risk-taking following the introduction of the TLAC rule.

To compare magnitudes across outcomes, we scale each treatment effect by the standard deviation of the respective outcome variable. The effects are economically meaningful. For the Z-score, whose standard deviation is 0.086, the treatment effect corresponds to a decrease of 0.29 standard deviations ($-0.236/0.8080 = -0.29$). By the same measure, the treatment effect is associated with increases of 0.18 standard deviations in the loan yield ($0.0060/0.0333=0.18$), 0.29 standard deviations in the provision ratio ($0.0030/0.0104=0.29$), and 0.24 standard deviations in the leverage ratio ($0.7910/3.3340=0.24$).

These magnitudes are comparable to those from the baseline panel regressions, in which a one-standard-deviation increase in the parent debt ratio is associated with changes of 0.13 to 0.23 standard deviations across the same outcomes. In the difference-in-differences analysis, the treatment effects range from 0.18 to 0.29 standard deviations, with signs opposite to the panel estimates, as expected given the decline in parent debt among Treat-BHCs. The similarity between the two sets of estimates suggests that the panel associations are not driven primarily by the endogeneity of parent debt.

Taken together, the evidence is consistent with a causal effect of parent debt on the risk-taking of subsidiary banks. As TLAC uncertainty was resolved in their favor, Treat-BHCs reduced their parent debt and their subsidiaries increased risk-taking. GSIBs, for which the resolution of TLAC scope carried no new information, exhibit no comparable change.

5 The impact of cross-guarantee on subsidiary bank’s risk-taking

Our previous analysis demonstrates that parent debt reduces risk-taking at the subsidiary bank level. We propose two complementary mechanisms to explain this finding.

The first mechanism operates through joint liability. When a parent company defaults, its subsidiary banks may bear liability for the parent’s obligations, effectively creating a contractual bail-in arrangement. This relationship is reciprocal: subsidiary bank-level distress can similarly propagate upward to the parent company.

The second mechanism stems from the cross-guarantee provisions of the Financial Institutions Reform, Recovery, and Enforcement Act of 1989 (FIRREA). Under FIRREA, the Federal Deposit Insurance Corporation (FDIC) is permitted to charge off any expected losses related to the failure of one subsidiary bank within a multi-bank holding company to the capital of affiliated subsidiary banks within the same organization ([Ashcraft \[2005\]](#)). We illustrate this with an example: consider two BHCs — A and B. BHC A operates three bank subsidiaries (A1, A2, and A3), whereas BHC B operates a single subsidiary (B1). If A1 defaults, the FDIC may require BHC A to deploy capital from A2 and A3 to absorb A1’s losses before any public intervention occurs. By contrast, if B1 defaults, BHC B has no affiliated subsidiaries from which to draw capital, necessitating an FDIC bailout. This asymmetry suggests that multi-bank BHCs face stronger internal discipline: because they must deploy their own capital to resolve a failing subsidiary, they have greater incentive to limit risk-taking ex ante.

Building on this, we empirically test the regulatory bail-in mechanism by examining whether BHCs with multiple bank subsidiaries curtail subsidiary banks’ risk-taking when

the parent company issues more external debt. For each bank subsidiary, we estimate the following model:

$$\begin{aligned}
RiskTaking_{ijt} = & \alpha + \beta_1 ParentDebtRatio_{j,t-4} + \beta_2 Multibank_{j,t} + \\
& + \beta_3 Multibank_{j,t} * ParentDebtRatio_{j,t-4} + \\
& + \gamma Control_{i,t-4} + \delta TobinQ_{j,t-4} + \lambda_i + \eta_t + \epsilon_{ijt}
\end{aligned} \tag{3}$$

where, for each bank subsidiary i affiliated with BHC j and in quarter t , *Multibank* equals to one if the BHC has more than one subsidiary banks, otherwise, zero. *Control* and *TobinQ* are the same as these we used in regression model (1). Table 2, Panel C, summarizes key variables used in the model.

Table 5 demonstrates a negative relationship between the cross-guarantee and subsidiary bank's risk-taking. For a one standard deviation increase in parent debt ratio (0.129), we observe an expected additional 21.66% decrease in the risk-weighted leverage (0.129 x -1.6793 x 100% = -21.66%) and a 2.53% decrease in the risk weight (0.129 x -0.1958 x 100% = -2.53%) if the BHC has more than one subsidiary banks.

Overall, the results presented in this section show that parent debt reduces subsidiary banks' risk-taking especially when there is more than one subsidiary bank within the BHC, potentially due to the cross-guarantee mechanism, which gives the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses.

6 Robustness test

The robustness check addresses concern regarding whether our results are truly driven by sample selection. In the difference-in-differences analysis, we use a sample from 2017 to 2022. For the robustness check, we limit our sample to this same period, that is, 2017–2022.

Table 2, Panel D, summarizes the sample statistics for the 2017–2022 period. On average, the parent debt ratio is 13.7%. The mean logarithm of asset size is 15.76. The average debt/equity ratio is 7.24, while the mean Tobin’s Q of BHCs is 1.04.

Table 6 shows a negative relationship between the parent debt ratio and the subsidiary bank’s risk-taking. Specifically, for a one standard deviation increase in the parent debt ratio (0.139), we observe an expected 5.08% increase in Z-Score ($0.3655 \times 0.139 \times 100\% = 5.08\%$), a 7.03% decrease in risk-weighted leverage ($-0.5062 \times 0.139 \times 100\% = -7.03\%$), and a 23.85% decrease in leverage ($-1.7158 \times 0.139 \times 100\% = -23.85\%$).

The robustness test confirms that a higher parent debt ratio reduces subsidiary banks’ risk-taking and that the main result is not driven by the sample selection.

7 A model of parent debt and subsidiary banks’ risk-taking

This section develops a theoretical framework to analyze the relationship between parent debt and the risk-taking behavior of subsidiary banks. Detailed derivations and proofs

are provided in the Appendix. The model operates over two periods, denoted as $t = 0$ and $t = 1$. We consider a BHC that wholly owns two distinct entities within its organizational structure: a subsidiary bank and a nonbank subsidiary.

7.1 Period $t = 0$

At $t = 0$, the consolidated balance sheet of the BHC comprises one unit of asset, D_c units of debt, and E_c units of equity. In addition to the consolidated balance sheet, the BHC maintains a parent company only balance sheet, which is subject to Federal Reserve oversight. The parent company holds E_p units of equity and issues D_p units of debt. It downstreams these funds to capitalize both the subsidiary bank and nonbank subsidiary proportionally. Specifically, the parent injects $\alpha(E_p + D_p)$ into the nonbank subsidiary and $(1 - \alpha)(E_p + D_p)$ into the subsidiary bank as equity investments, where α and $1 - \alpha$ represent the relative asset sizes of the nonbank subsidiary and subsidiary bank, respectively.

Under this formulation,

$$E_p + D_p \tag{1}$$

represent the uniform equity capital ratio for both subsidiaries, as capitalized by the parent company. While capital allocation within a BHC is typically endogenous, we treat the subsidiary capital ratios as fixed parameters. This simplifying assumption can be motivated by the presence of binding regulatory capital requirements at both the subsidiary bank and the BHC levels. The nonbank subsidiary's assets, debt, and equity are denoted by α , D_n , and E_n , respectively. Similarly, the subsidiary bank's assets, debt, and equity are denoted by $1 - \alpha$, D_b , and E_b , respectively. The balance sheets for the consolidated BHC, the parent company, the nonbank subsidiary, and the subsidiary

bank are summarized in Table 1. By construction, E_b and E_n constitute the assets of the parent company (A_b and A_n).

Table 1: Balance Sheets at $t = 0$

Consolidated BHC	Parent Company	Nonbank Subsidiary	Subsidiary Bank
$\begin{array}{c c} 1 & D_c \\ \hline & E_c \end{array}$	$\begin{array}{c c} A_b & E_p \\ \hline A_n & D_p \end{array}$	$\begin{array}{c c} \alpha & E_n \\ \hline & D_n \end{array}$	$\begin{array}{c c} 1 - \alpha & E_b \\ \hline & D_b \end{array}$

At $t = 0$, the subsidiary bank encounters an investment opportunity that yields a payoff R_b with probability $P(R_b)$ and a salvage value S with probability $1 - P(R_b)$. To capture a classical risk-return trade-off, we assume the probability of success is a decreasing, linear function of the target payoff:

$$P(R_b) = 1 - \beta R_b, \quad (2)$$

implying that higher returns are inherently less likely. The payoff distribution of the subsidiary bank's investment is therefore:

$$R = \begin{cases} R_b, & \text{with probability } 1 - \beta R_b, \\ S, & \text{with probability } \beta R_b. \end{cases}$$

To ensure that the limited liability friction is economically meaningful, we impose the following condition on the magnitude of the salvage value:

$$S < 1 - E_p. \quad (3)$$

This condition guarantees that, in the absence of capitalization financed by parent debt ($D_p = 0$), the potential loss from the investment is sufficiently severe to render the subsidiary bank insolvent; that is, $S(1 - \alpha) < (1 - \alpha)(1 - E_p)$. If we were to assume the

alternative ($S \geq 1 - E_p$), the equity capital endogenously supplied by the BHC would always be abundant enough to absorb the investment losses, preventing subsidiary bank bankruptcy and neutralizing the core risk-shifting incentives.

The nonbank subsidiary generates a return of R_n . For simplicity, we assume the nonbank activities yield a risk-free return, denoted as r_f . This assumption reflects the reality that many nonbank operations—such as advisory services, asset management, and securities underwriting—generate stable fee income that lacks the default risk typical of bank lending portfolios. Normalizing the risk-free rate to 1, and assuming the subsidiary bank does not exercise deposit market power in its deposit rate setting, we establish:

$$r_b = r_f = 1. \quad (4)$$

This provides a neutral benchmark where the subsidiary bank cannot extract rents on the liability side.

7.2 Period $t = 1$

At $t = 1$, the realized balance sheets for the subsidiary bank are determined by the outcome of the subsidiary bank's investment. The balance sheets for the nonbank subsidiary, and the subsidiary bank are summarized in Table 2.

Table 2: Subsidiaries' Balance Sheets at $t = 1$

Nonbank Subsidiary	Subsidiary Bank (Success)	Subsidiary Bank
$R_n \alpha \mid \begin{array}{c} R_n \alpha - D_n \\ D_n \end{array}$	$R_b(1 - \alpha) \mid \begin{array}{c} R_b(1 - \alpha) - D_b \\ D_b \end{array}$	$S(1 - \alpha) \mid \begin{array}{c} \max\{S(1 - \alpha) - D_b, 0\} \\ D_b \end{array}$

The consolidated BHC shareholder chooses the risk profile R_b to maximize the expected payoff. The objective function is given by:

$$\begin{aligned} \max_{R_b} \quad \mathbb{E}[L] = & P(R_b) [R_n \alpha - D_n + R_b(1 - \alpha) - D_b - D_p] \\ & + (1 - P(R_b)) [R_n \alpha - D_n + \max\{S(1 - \alpha) - D_b, 0\} - D_p], \end{aligned} \quad (5)$$

where the debt obligations of the two subsidiaries are strictly determined by the uniform parent equity allocation:

$$D_b = (1 - \alpha)(1 - E_p - D_p), \quad (6)$$

$$D_n = \alpha(1 - E_p - D_p). \quad (7)$$

Before analyzing the equilibrium under the limited liability distortion, it is instructive to establish the first-best investment policy. The first-best risk level is determined by maximizing the expected investment payoff of the subsidiary bank's production technology, without regard to the BHC's capital structure or moral hazard. Taking the first-order derivative of the expected payoff with respect to R_b yields the first-best solution:

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \quad (8)$$

The subsidiary bank fails if and only if its equity value falls below its debt obligation, $S(1 - \alpha) < D_b$. Substituting Equation (6), this condition simplifies to:

$$D_p < (1 - E_p) - S \equiv \hat{D}'_p. \quad (9)$$

Furthermore, when the subsidiary bank fails, the parent/BHC survives if and only if the residual equity value of the nonbank subsidiary is sufficient to cover the parent's independent debt obligation D_p . This survival condition is $R_n\alpha - D_n - D_p \geq 0$, which, applying Equation (7), reduces to:

$$D_p \leq \frac{\alpha[R_n - (1 - E_p)]}{1 - \alpha} \equiv \widehat{D}_p'' \quad (10)$$

Building on these thresholds, the subsequent analysis examines the BHC's optimal risk-taking decision across the two broad cases (1) no parent/BHC bankruptcy when the subsidiary bank fails, and (2) parent/BHC fails. When the parent/BHC fails, the subsidiary bank can fail or not.

7.3 Case 1: No parent/BHC bankruptcy ($D_p \leq \widehat{D}_p''$)

We begin our equilibrium analysis with the parameter space where $D_p \leq \widehat{D}_p''$. In this regime, the BHC possesses sufficient nonbank equity to fully absorb the parent debt obligation, ensuring it survives even if the subsidiary bank fails on its investment.

Lemma 1. *If and only if $\alpha > \frac{(1-E_p)-S}{R_n-S} \equiv \widehat{\alpha}$, it holds that $\widehat{D}_p' < \widehat{D}_p''$. Excessive risk-taking occurs when $D_p < \widehat{D}_p'$, whereas the first-best risk level emerges when $D_p \in [\widehat{D}_p', \widehat{D}_p'']$.*

See Appendix A1: Proof of Lemma 1.

When $D_p < \widehat{D}_p'$, the subsidiary bank fails under distress (realizing salvage value S), but the BHC survives. Limited liability protection at the subsidiary bank level distorts

the risk-taking incentives of the BHC shareholder. Mathematically, solving the FOC of the objective function yields the optimal risk level:

$$R_b^* = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}. \quad (11)$$

Because the default condition ensures $1 - E_p - D_p > S$, it strictly follows that $R_b^* > R_b^{FB}$, confirming that excessive risk-taking occurs.

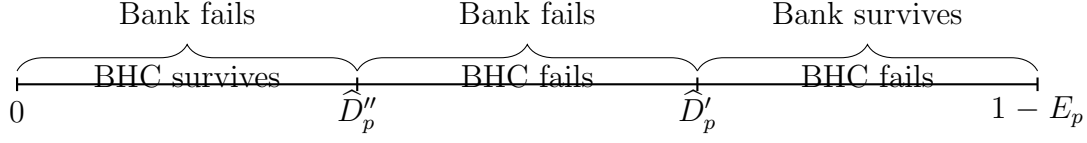
In contrast, when $D_p \in [\hat{D}_p', \hat{D}_p'']$, both the subsidiary bank and the parent/BHC survive the bad state. The losses are fully passed onto the shareholder of the BHC. The absence of limited liability distortion on the subsidiary bank level fully aligns the private risk-taking decision with the first-best level, yielding the first-best outcome, R_b^{FB} .

Lemma 2. *A more significant nonbank subsidiary, i.e., an increase in α (given $\alpha > \hat{\alpha}$), reduces risk-taking for a given D_p .*

An increase in α acts to increase the parent survival threshold $\hat{D}_p''$, while the subsidiary bank failure threshold $\hat{D}_p'$ remains unchanged. This cleanly expands the parameter interval $[\hat{D}_p', \hat{D}_p'']$ governed by the first-best risk level. Intuitively, when α approaches 0 and the equity value from the nonbank subsidiary becomes negligible, the limited liability distortion is fully transmitted onto the parent/BHC level, creating excessive risk-taking incentives.

Lemma 3. *If and only if $\alpha < \frac{(1-E_p)-S}{R_n-S} \equiv \hat{\alpha}$, it holds that $\hat{D}_p' > \hat{D}_p''$. Excessive risk-taking occurs when $D_p < \hat{D}_p''$ if the parent/BHC does not fail.*

When α falls below $\hat{\alpha}$, the ordering of the critical thresholds flips. The possible outcome boundaries are visually summarized in the figure below:



7.4 Case 2: Parent/BHC bankruptcy risk ($D_p > \hat{D}_p''$)

We now shift to the parameter space where $D_p > \hat{D}_p''$, exposing the parent company to severe bankruptcy risk. When the parent/BHC fails, its shareholders receive a terminal equity payoff of exactly zero, tightly shielded by parent-level limited liability.

7.4.1 Both the subsidiary bank and the parent/BHC fail ($\hat{D}_p'' < D_p < \hat{D}_p'$)

Lemma 4. *When both the parent/BHC and subsidiary bank fail, excessive risk-taking still occurs because limited liability distortion emerges at the subsidiary bank level.*

See Appendix A2: Proof of Lemma 4.

Mathematically, the optimal risk level chosen by the BHC shareholder in this regime is given by:

$$R_b = \frac{1 + \beta(1 - E_p - D_p)}{2\beta}. \quad (12)$$

The subsidiary bank still takes excessive risk because the limited liability constraint binds strictly at the subsidiary bank level.

7.4.2 The parent/BHC fails, but the subsidiary bank survives ($D_p > \hat{D}'_p$)

Lemma 5. *When the parent/BHC fails but the subsidiary bank remains solvent, it yields the first best solution with*

$$R_b^{FB} = \frac{1 + \beta S}{2\beta}. \quad (13)$$

See Appendix A3: Proof of Lemma 5.

When the parent/BHC fails but the subsidiary bank remains solvent, the absence of a limited liability distortion at the subsidiary bank level aligns the private risk-taking decision directly with the socially optimal first-best level.

8 Conclusion

This paper provides evidence on how parent debt affects subsidiary banks' risk-taking. Using a comprehensive dataset covering all domestic U.S. commercial banks and their corresponding BHCs from 2012 to 2022, combined with a difference-in-differences analysis using the COVID-19 outbreak as a natural experiment, we find that increased parent debt is associated with reduced risk-taking by subsidiary banks. This effect is particularly prominent in BHCs with significant nonbank business or more than one bank subsidiary, which would give the parent company greater incentives to constrain risk-taking in the subsidiary bank to prevent any spillover losses. These findings highlight the need to consider the capital structures of the subsidiary, the parent, and the overall BHC—as well as the organizational structure—when assessing a subsidiary bank's risk profile.

References

- Viral V Acharya, Lasse H Pedersen, Thomas Philippon, and Matthew Richardson. Measuring systemic risk. *The review of financial studies*, 30(1):2–47, 2017.
- Jonathan Acosta-Smith, Michael Grill, and Jan Hannes Lang. The leverage ratio, risk-taking and bank stability. *Journal of Financial Stability*, page 100833, 2020.
- Adam B Ashcraft. Are banks really special? new evidence from the fdic-induced failure of healthy banks. *American Economic Review*, 95(5):1712–1730, 2005.
- Dafna Avraham, Patricia Selvaggi, and James I Vickery. A structural view of us bank holding companies. *Economic Policy Review*, 18(2):65–81, 2012.
- Marianne Bertrand, Esther Duflo, and Sendhil Mullainathan. How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275, 2004.
- John H Boyd and David E Runkle. Size and performance of banking firms: Testing the predictions of theory. *Journal of Monetary Economics*, 31(1):47–67, 1993.
- Nicola Cetorelli and Linda S Goldberg. Banking globalization and monetary transmission. *The Journal of Finance*, 67(5):1811–1843, 2012.
- Rebecca S Demsetz and Philip E Strahan. Diversification, size, and risk at bank holding companies. *Journal of Money, Credit, and Banking*, pages 300–313, 1997.
- Andrew Ellul and Vijay Yerramilli. Stronger risk controls, lower risk: Evidence from us bank holding companies. *The Journal of Finance*, 68(5):1757–1803, 2013.

- Xavier Freixas, Gyöngyi Lóránth, and Alan D Morrison. Regulating financial conglomerates. *Journal of Financial Intermediation*, 16(4):479–514, 2007.
- Beverly Hirtle, Anna Kovner, and Matthew Plosser. The impact of supervision on bank performance. *The Journal of Finance*, 75(5):2765–2808, 2020.
- Michael C Jensen and William H Meckling. Theory of the firm. *Managerial behavior, agency costs and ownership structure*, 3(4):305–360, 1976.
- Charles Kahn and Andrew Winton. Moral hazard and optimal subsidiary structure for financial institutions. *The Journal of Finance*, 59(6):2531–2575, 2004.
- Michael C Keeley. Deposit insurance, risk, and market power in banking. *The American Economic Review*, pages 1183–1200, 1990.
- Luc Laeven and Ross Levine. Is there a diversification discount in financial conglomerates? *Journal of Financial Economics*, 85(2):331–367, 2007.
- Zhao Li, Kebin Ma, and Lucy White. The making of (modern) banks. *WBS Finance Group Research Paper*, 2024.
- Elisa Luciano and Clas Wihlborg. Financial synergies and systemic risk in the organization of bank affiliates. *Journal of Banking & Finance*, 88:208–224, 2018.
- Ronald E Shrieves and Drew Dahl. The relationship between risk and capital in commercial banks. *Journal of Banking & Finance*, 16(2):439–457, 1992.
- Jennifer W Tucker, Ying Zhou, and Jigao Zhu. Parent-only balance sheet information and credit risk assessments. *Accounting Horizons*, pages 1–22, 2023.

Table 1: Variable definitions

Variable	Definition
pdratio_ultp	pdratio_ultp is the ratio of the immediate parent's non-deposit debt liabilities to the BHC's non-deposit liabilities, measured quarterly as: (BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / (BHCK2948 (group-level consolidated total liabilities) – BHDM6631 (group-level consolidated noninterest-bearing deposits in domestic offices) – BHDM6636 (group-level consolidated interest-bearing deposits in domestic offices) – BHFN6631 (group-level consolidated noninterest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs) – BHFN6636 (group-level consolidated interest-bearing deposits in foreign offices, Edge and Agreement subsidiaries, and IBFs))
pdratioa_ultp	pdratioa_ultp is the ratio of the immediate parent's non-deposit debt liabilities to the BHC's assets, measured quarterly as: (BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / BHCK2170 (BHC's total assets)
pdratiowda_ultp	pdratiowda_ultp is the ratio of the immediate parent's total liabilities to the BHC's assets, measured quarterly as: (BHCP2200 (parent company's deposits) + BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / BHCK2170 (BHC's total assets)
loan_yield	loan_yield is the ratio of the subsidiary bank's total interest income from loans to its total loans: RIAD4010 (total interest and fee income on loans) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or RIAD4010 (total interest and fee income on loans) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041
CI_ratio	CI_ratio is the ratio of commercial and industrial loans to total loans: (RCFD1763 (commercial and industrial loans to U.S. addressees) + RCFD1764 (commercial and industrial loans to non-U.S. addressees)) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or (RCON1763 (commercial and industrial loans to U.S. addressees) + RCON1764 (commercial and industrial loans to non-U.S. addressees)) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041
provision_ratio	provision_ratio is the ratio of provisions to total loans: RIAD4230 (provisions for credit losses) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or RIAD4230 (provisions for credit losses) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041
lev_bsub	lev_bsub is the subsidiary bank's debt-to-equity ratio: RCON2948 (bank subsidiary's total liabilities, FFIEC 041) / (RCON2170 (bank subsidiary's total assets, FFIEC 041) – RCON2948 (bank subsidiary's total liabilities, FFIEC 041)), or RCFD2948 (bank subsidiary's total liabilities, FFIEC 031) / (RCFD2170 (bank subsidiary's total assets, FFIEC 031) – RCFD2948 (bank subsidiary's total liabilities, FFIEC 031))

Zscore	Zscore is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as total equity (RCFD3210, total bank equity capital at the end of the current period) divided by total assets (RCFD2170) for FFIEC 031, or total equity (RCON3210, total bank equity capital at the end of the current period) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Zscore_nrw	Zscore_nrw is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as book equity (RCFD3632 (retained earnings) + RCFD3839 (surplus, excluding all surplus related to preferred stock) + RCFD3230 (common stock)) divided by total assets (RCFD2170) for FFIEC 031, or book equity (RCON3632 (retained earnings) + RCON3839 (surplus, excluding all surplus related to preferred stock) + RCON3230 (common stock)) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
ln_ta_bsub	ln_ta_bsub is the logarithm of the subsidiary bank's total assets: $\ln(\text{RCON2170 (bank subsidiary's total assets, FFIEC 041)})$, or $\ln(\text{RCFD2170 (bank subsidiary's total assets, FFIEC 031)})$
HQLA_ratio	HQLA_ratio is the ratio of the BHC's high-quality liquid assets to the BHC's total assets: $(\text{BHCM3531 (U.S. Treasury securities)} + \text{BHCM3532 (U.S. government agency obligations, excluding mortgage-backed securities)} + \text{BHCKG379 (residential pass-through MBS issued or guaranteed by FNMA, FHLMC, or GNMA)} + \text{BHCKG380 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, including CMOs, REMICs, and stripped MBS)} + \text{BHCKK197 (commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies)} + \text{BHCKG301 (residential pass-through securities guaranteed by GNMA, held to maturity)} + \text{BHCKG303 (residential pass-through securities guaranteed by GNMA, available for sale)} + \text{BHCKG305 (residential pass-through securities issued by FNMA and FHLMC, held to maturity)} + \text{BHCKG307 (residential pass-through securities issued by FNMA and FHLMC, available for sale)} + \text{BHCKKX53 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, held to maturity)} + \text{BHCKKX55 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, available for sale)} + \text{BHCKG313 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity)} + \text{BHCKG315 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale)} + \text{BHCKK143 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, held to maturity)} + \text{BHCKK145 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, available for sale)} + \text{BHCKK151 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity)} + \text{BHCKK153 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale)} + \text{BHCKHT51 (U.S. government agency and sponsored agency obligations, held to maturity)} + \text{BHCKHT53 (U.S. government agency and sponsored agency obligations, available for sale)} + \text{RCFD0090 (balances due from Federal Reserve Banks)} \text{ for FFIEC 031, or } \text{RCON0090 (balances due from Federal Reserve Banks)} \text{ for FFIEC 041} + \text{BHCK0213 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, held to maturity)} + \text{BHCK1287 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, available for sale)}) / \text{BHCK2170 (BHC's total assets)}$

tobinq_ultp

Tobin's Q is the group-level consolidated Tobin's Q, defined as the ratio of the market value of the BHC (MV) to the replacement cost of its assets (RC). It is estimated as $\text{tobinq_ultp} = (\text{AT} + \text{ME} - \text{BE}) / \text{AT}$, where AT is total assets, ME is the market value of equity at year end (stock price multiplied by common shares outstanding), and BE is book equity (stockholders' equity + deferred taxes + investment tax credit – preferred stock). Calculated using the WRDS code; see link for details: <https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/>

Table 2: summary statistics for general regression

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
loan_yield	8,907	0.0482	0.0157	0.0372	0.0415	0.0459	0.0509	0.0571	0.0184	0.1364
provision_ratio	8,907	0.0032	0.0068	-0.0011	0.0002	0.0016	0.0037	0.0087	-0.0099	0.0399
CI_ratio	8,572	0.2002	0.1285	0.0588	0.1027	0.1760	0.2776	0.3902	0.0000	0.5655
zscore_annual	9,871	3.6525	0.5889	3.0928	3.3643	3.6045	3.8726	4.2256	2.0747	6.2843
zscore_nrw_annual	9,871	3.6545	0.5785	3.0986	3.3667	3.6063	3.8760	4.2357	2.1379	6.1054
lev_bsub	9,871	7.2952	3.0585	0.8505	6.3290	7.7471	9.0639	10.3169	0.0076	14.2192
pdratioa_ultp	9,783	0.0187	0.0328	0.0000	0.0000	0.0084	0.0195	0.0542	0.0000	0.3356
tobinq_ultp	9,871	1.0320	0.0539	0.9797	1.0000	1.0230	1.0532	1.0877	0.9122	1.5686
ln_ta_bsub	9,871	15.0292	2.2805	12.5615	13.9251	14.9999	16.3090	17.7974	7.2167	21.8888
HQLA_ratio	9,783	0.1543	0.0812	0.0663	0.0942	0.1390	0.1994	0.2607	0.0025	0.5890

[illegible]

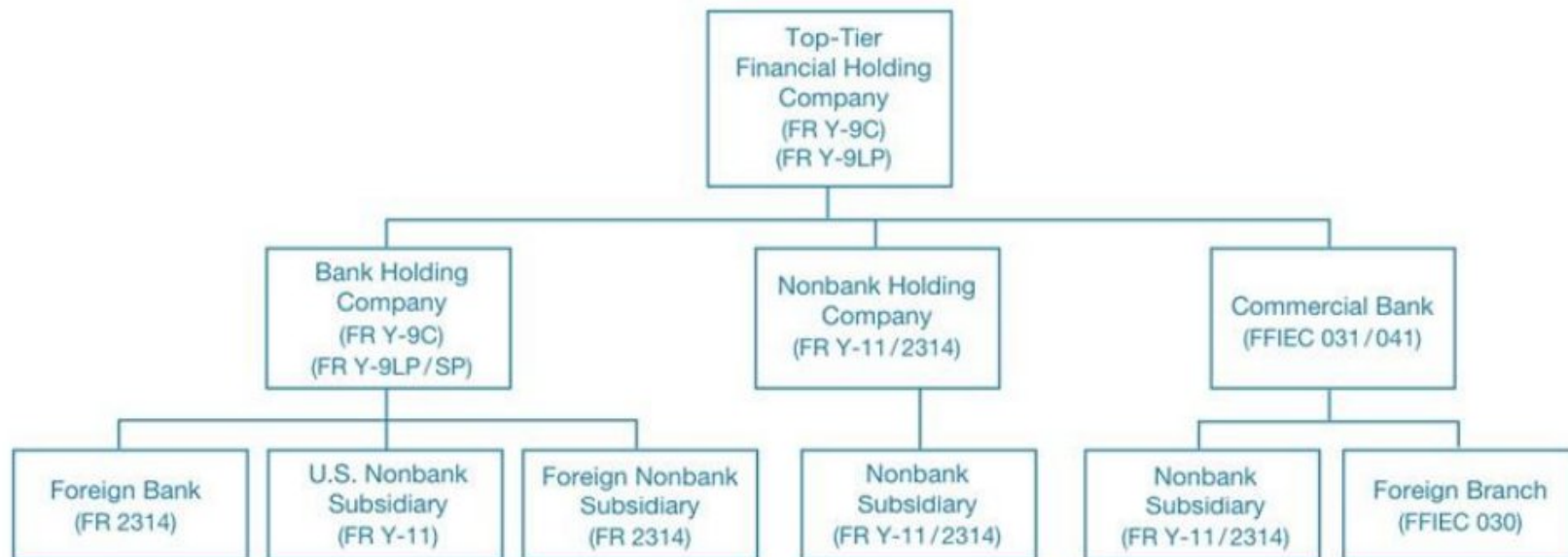
Table 4: summary statistics for large BHCs, and the sample is collapsed

VARIABLES	N	2012-2103			N	2016-2017		
		mean	sd	p50		mean	sd	p50
HQLA_ratio	36	0.155	0.0440	0.156	36	0.206	0.0472	0.211
zscore_post	34	3.676	0.750	3.638	34	3.536	0.881	3.569
zscore_nrw_post	26	3.666	0.808	3.653	26	3.606	0.722	3.591
provision_ratio	20	0.00801	0.0104	0.00324	20	0.00951	0.0150	0.00305
CI_ratio	19	0.234	0.163	0.221	19	0.255	0.170	0.271
loan_lease_yield	20	0.0543	0.0333	0.0435	20	0.0533	0.0359	0.0403
lev_bsub	26	4.984	3.334	6.386	26	5.491	3.559	6.788
tobinq_ultp	35	1.048	0.0913	1.014	35	1.061	0.0700	1.042
ln_ta_bsub	36	15.15	3.862	16.17	36	15.33	4.113	16.98
pdratioa_ultp	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0277
pdratiowda_ultp	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0278
pdratio_ultp	36	0.162	0.113	0.182	36	0.164	0.0774	0.189

[illegible]

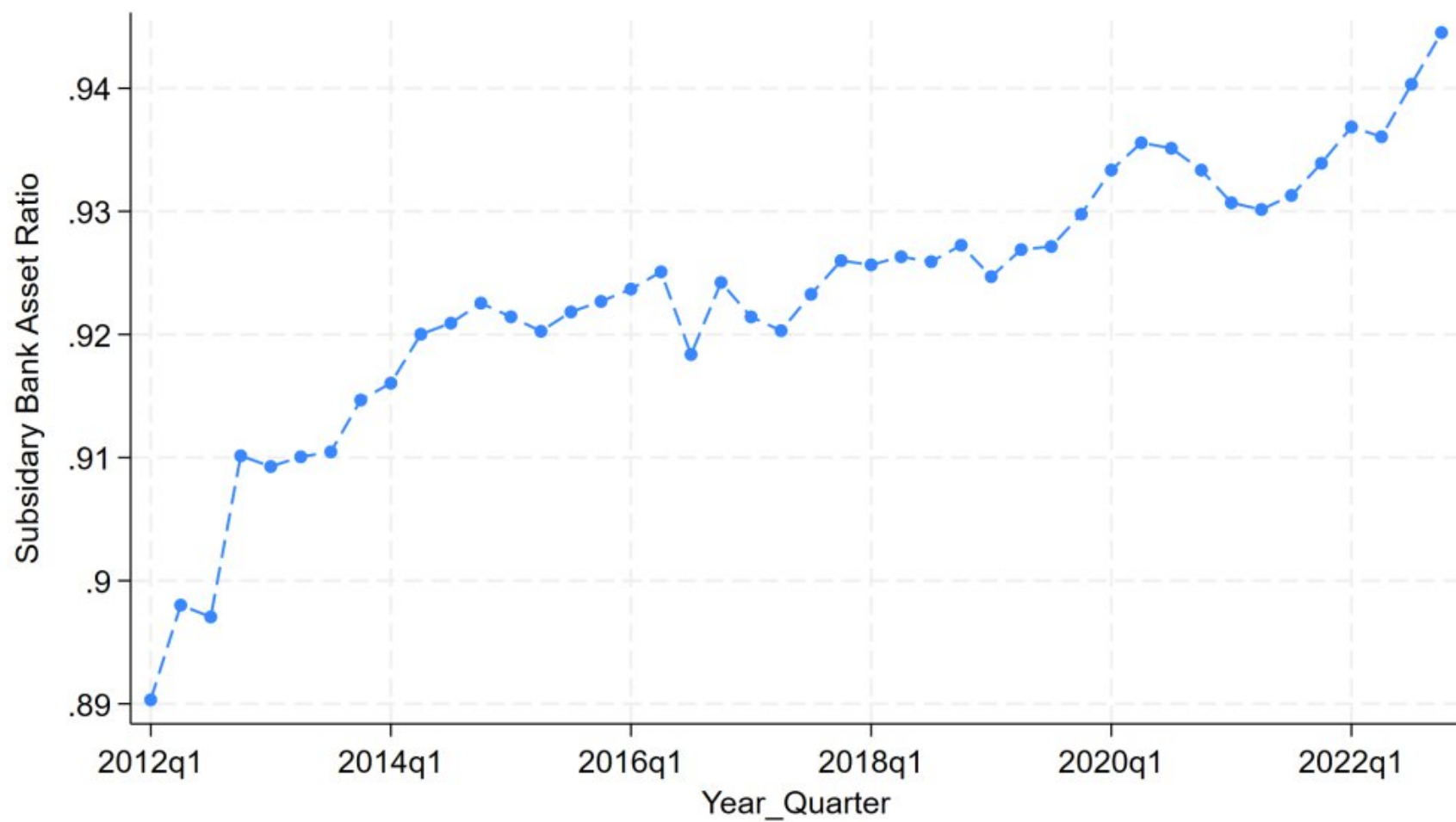
Figure 1: The organizational structure of a large BHC

Stylized Structure of a Large Bank Holding Company



Data sources: Avraham et al. (2012)

Figure 2: Subsidiary bank's asset ratio over time



Source: FR Y-9C and Call Reports

Figure 3: Parallel trend for large non-GSIB BHCs and the control group

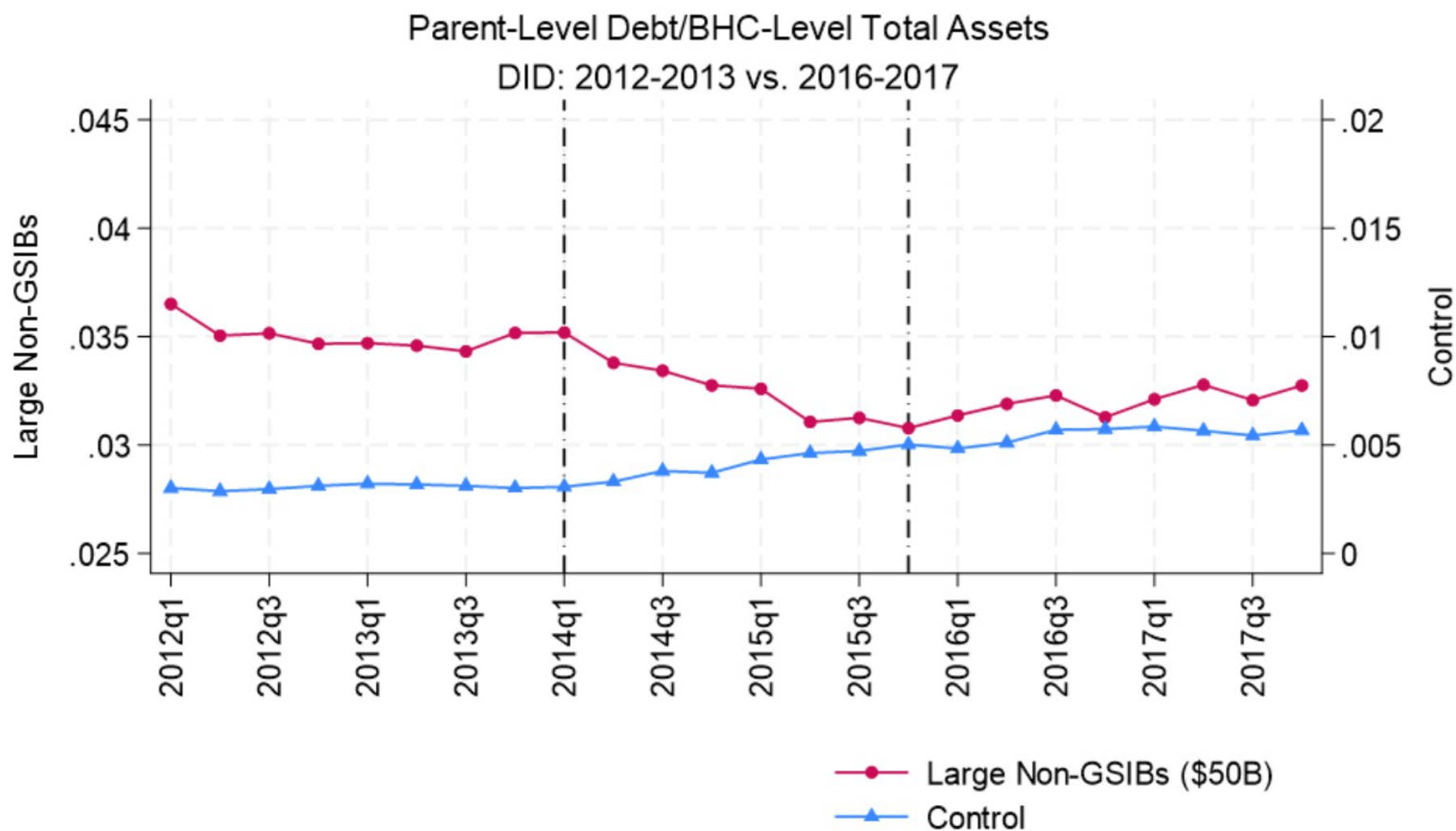


Figure 4: Parallel trend for GSIBs and the control group

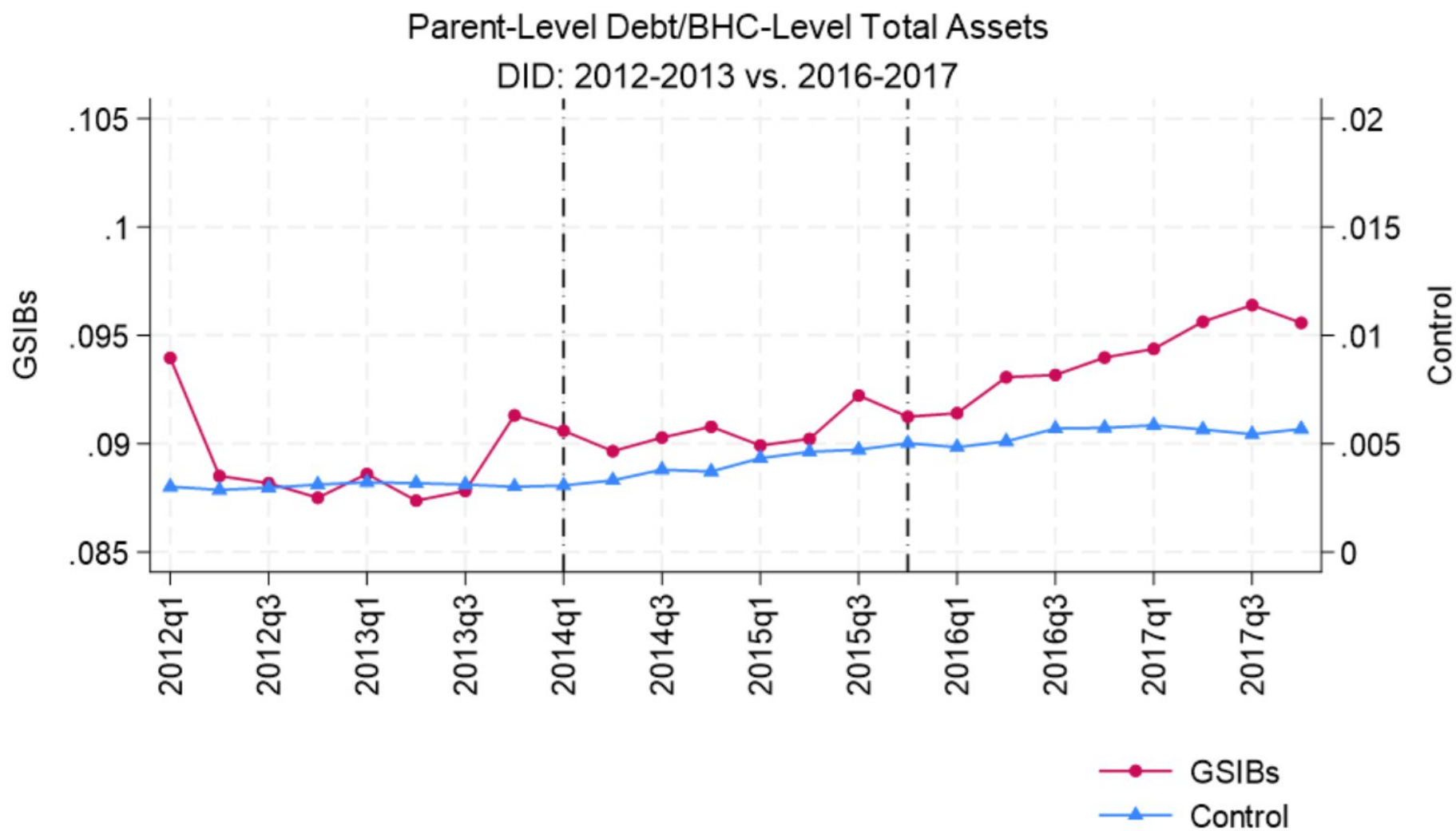


Figure 5: Parallel trend for GSIBs and the control group

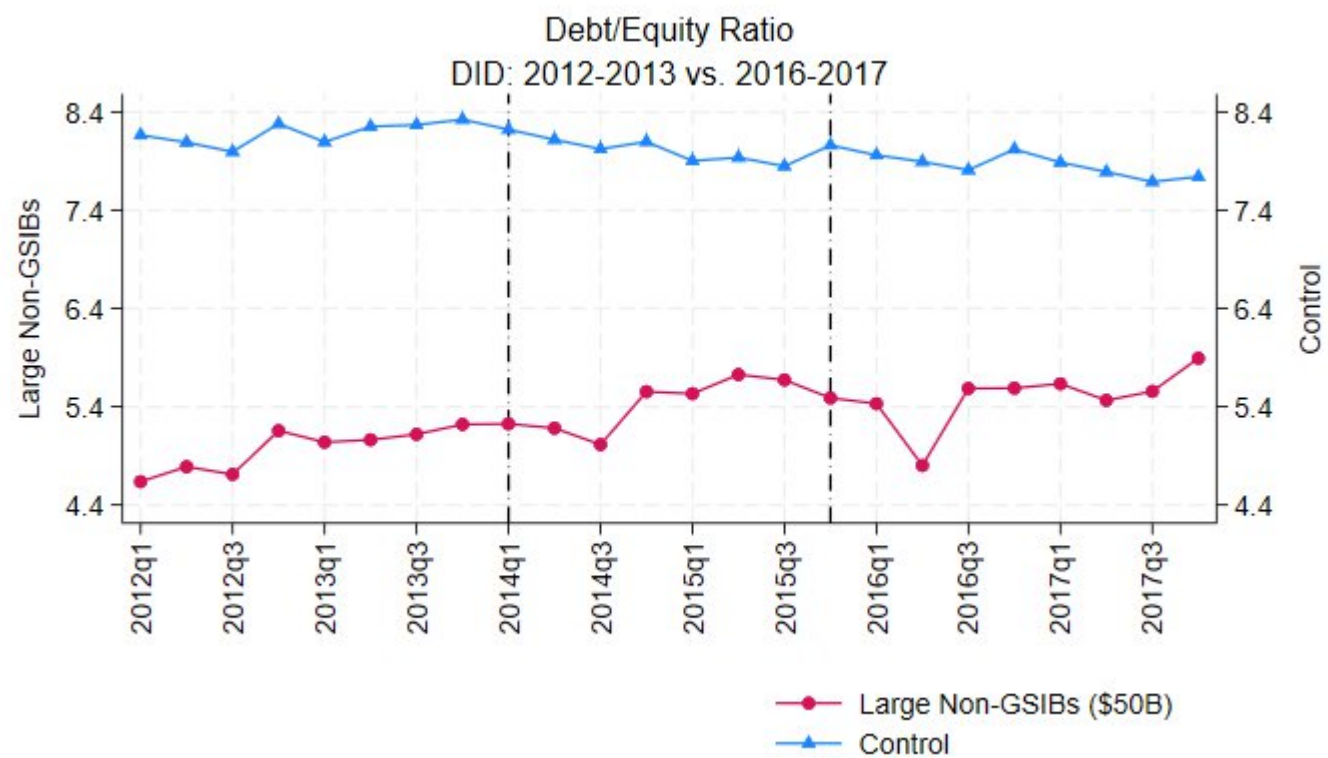


Figure 5: Parallel trend for GSIBs and the control group

